# Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Agentic AI · Full-Stack Data Services · MCP · Distributed Systems · Cloud-Native Pipelines · Python · TypeScript · AWS · Production Reliability · AI-Assisted Development

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## PROFESSIONAL SUMMARY

Software engineer with 8 years delivering scalable full-stack applications and data services in enterprise SaaS - and two production agentic AI systems that map directly to CEAI's mission of AI-native commercial platforms. At Entrata, owned the HUD income-limit data pipeline end-to-end for 7 years (federal source → 180,000 domain records powering eligibility decisions for 100K+ monthly active users): distributed data services, event-driven async processing with retry and idempotency guarantees, cloud-native infrastructure (AWS S3, EC2, RDS), and full observability stack - 99.5% uptime SLA. LeaseI is an 8-agent LangGraph agentic platform with multi-source RAG, deterministic guardrails, HITL escalation, and MCP-compatible tool interfaces - 93.6% precision across 1,240 compliance test cases. AgentProbe is an open-source LLM evaluation framework: 5-metric quality engine, LLM-as-judge scoring, CI-integrated regression gate, multi-provider routing. Uses AI-assisted development tools (Claude Code, MCP) daily in professional engineering workflows. 300+ code reviews, 5+ engineers mentored, full-stack delivery across Python, TypeScript, JavaScript, and FastAPI services. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**AI & Agentic Systems:** Agentic AI, LangGraph, LangChain, Multi-Agent Orchestration, MCP (Model Context Protocol), RAG Pipelines, LLM Evaluation, LLM-as-Judge, CI-Integrated Regression Gating, HITL, Guardrails, Structured Outputs, Prompt Engineering, ChromaDB, Vector Search, Embeddings, Anthropic API (Claude), OpenAI API, Ollama, Claude Code, AI-Assisted Development, BERT, spaCy, NLP, OCR

**Full-Stack & Distributed Systems:** Python (8yr), TypeScript, JavaScript, FastAPI, REST APIs, SQL, PHP, Bash, Docker, Microservices, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue - cloud-native comparable to Azure), RabbitMQ (event-driven async, retry, idempotency), Redis, PostgreSQL, MySQL, Oracle, NoSQL, PySpark, Apache Spark, Linux, Git

**Reliability & Data Engineering:** Observability, Production Monitoring, Performance Engineering, Data Pipelines, ETL, Data Modeling, Batch Processing, Streaming-Compatible Async, Privacy & Security Engineering, PHI Compliance (Libsodium), Feature Engineering, PyTorch, scikit-learn, NumPy, Pandas, Tableau, Agile/Scrum

## EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Built full-stack agentic AI application (LangGraph, FastAPI, Python) for mission planning - multi-agent orchestration with MCP-compatible tool interfaces, RAG retrieval over distributed data sources, deterministic guardrails on outputs, CI/CD-integrated quality gate at every release.
- Used AI-assisted development tools (Claude Code) throughout the engineering lifecycle; implemented HITL escalation and observability on agent performance, retrieval latency, and reliability.

### George Mason University

Aug 2024 - May 2026

**Graduate Teaching Assistant & Research Associate**

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering and statistical validation across 190+ countries.
- Mentored 600+ students in Python, SQL, distributed systems, and data engineering; designed monitoring dashboards improving research comprehension by 40%.

### Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **Full-stack data services and APIs at production scale** - 20+ scalable microservice modules at 99.5% uptime across 100K+ monthly active users; secure REST APIs and distributed data services (Python, JavaScript, TypeScript, FastAPI, PHP); AWS cloud-native infrastructure (S3, EC2, RDS, Glue); Docker containerization (50% scalability improvement, 30% deployment reduction).
- **Large-scale data pipelines and batch processing** - owned HUD income-limit pipeline 7 years (federal source → 180,000 domain records); RabbitMQ event-driven async processing with retry, idempotency, and dead-letter guarantees; Redis caching; 200+ SQL/PostgreSQL/Oracle query optimizations (30% performance gain); PySpark batch processing at scale.
- **AI-assisted development, observability, and reliability engineering** - AI-driven observability and monitoring tooling reducing triage time 40%; leveraged AI coding assistants to accelerate development and maintain quality; 100% PHI compliance (Libsodium-encrypted pipelines); sub-10 bug count maintained for 6+ consecutive months across multi-tenant production environments.
- **Technical leadership across engineering teams** - 300+ peer code reviews enforcing engineering standards and secure coding practices; mentored 5+ engineers on distributed systems, data pipelines, and AI best practices; CI/CD via GitHub Actions; collaborated with product, data, and compliance stakeholders to translate ambiguous business requirements into reliable technical solutions.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built backend modules (PHP, Python, PostgreSQL, NoSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

## PROJECTS

### Leasel: 8-Agent Agentic Platform with MCP & RAG

Agentic · MCP · RAG

Python · LangGraph (8-agent) · FastAPI · MCP · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_raq

Production agentic AI platform: 8-agent LangGraph stateful workflow with MCP-compatible tool interfaces, multi-source RAG retrieval, deterministic guardrails on regulated outputs, HITL escalation, and source-linked answers with 100% citation correctness. Full observability on agent performance, retrieval latency, and reliability. 93.6% precision, 87.9% F1 across 1,240 compliance test cases.

### AgentProbe: Open-Source LLM Evaluation Framework

Evals · CI/CD · Observability

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation and observability platform: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, LLM-as-judge scoring, multi-provider routing. Built with AI-assisted development tools throughout - zero regressions shipped across 1,240+ test cases.

### AI on FHIR: Full-Stack NLP-Driven Query System

Full-Stack · NLP · APIs

Python · Flask · LLMs · FHIR · React · TypeScript · Docker · spaCy · BERT | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application (Python backend + React/TypeScript frontend) translating natural-language queries into FHIR API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured records.

## EDUCATION

**M.S. Data Analytics Engineering -** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering -** JNTU Anantapur — 2013-2017